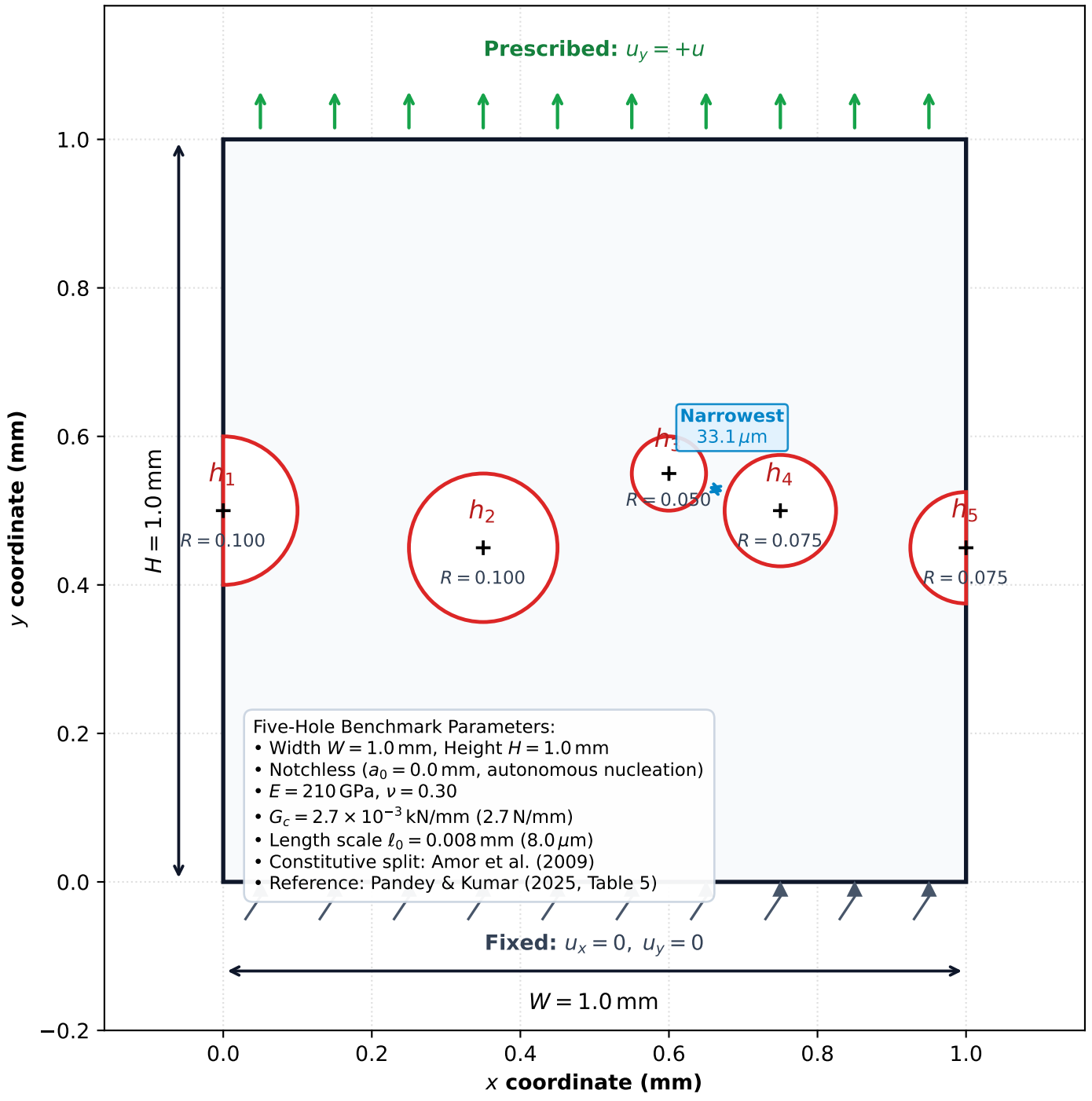
Pandey & Kumar (2025) Section 4.5 Five-Hole Benchmark Geometry
(Reproducing Figure 22(a) Specification Without Initial Notch)
Prescribed: $u_y = +u$
$H = 1.0\,\text{mm}$
h_1
$R = 0.100$
h_2
$R = 0.100$
Narrowest
33.1 μm
h_3
$R = 0.050$
h_4
$R = 0.075$
h_5
$R = 0.075$
Five-Hole Benchmark Parameters:
• Width $W = 1.0\,\text{mm}$, Height $H = 1.0\,\text{mm}$
• Notchless ($a_0 = 0.0\,\text{mm}$, autonomous nucleation)
• $E = 210\,\text{GPa}$, $\nu = 0.30$
• $G_c = 2.7 \times 10^{-3}\,\text{kN/mm}$ (2.7 N/mm)
• Length scale $\ell_0 = 0.008\,\text{mm}$ (8.0 μm)
• Constitutive split: Amor et al. (2009)
• Reference: Pandey & Kumar (2025, Table 5)
Fixed: $u_x = 0,\ u_y = 0$
$W = 1.0\,\text{mm}$
y coordinate (mm)
x coordinate (mm)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0